

SMRITI: An Evidence-Aware, Contradiction-Constrained Knowledge Graph Architecture for Heterogeneous Notes

Abhijnan B C
p9713889@gmail.com

Abstract

We present SMRITI, a system that converts unstructured personal note vaults into a reliability-scored, contradiction-aware knowledge graph by combining claim extraction, embedding-retrieval candidate generation, NLI-based relationship classification, signed-graph contradiction partitioning, and multi-signal evidence fusion. An earlier version of this system reported impressive-looking self-evaluation numbers that turned out, on inspection, to be structural proxies rather than semantic evaluation (e.g. treating claim “precision” as the fraction of extracted claims that are simply non-empty) and a contradiction classifier whose measured precision was 3.2%. Rather than polish those numbers, we rebuilt the evaluation layer and then rebuilt the algorithm the old evaluation was hiding problems in. This paper reports that reconstruction: (i) we deleted every fabricated metric from the system’s self-certification layer and replaced it with a frozen, five-claim research registry in which a claim without real supporting evidence is honestly reported as *not evaluable*, never defaulted to a passing score; (ii) we found and fixed the root algorithmic cause of the 3.2% contradiction precision — a resolver that checked a candidate relationship against only one alternative class instead of requiring it to be the genuine arg-max among support/contradiction/neutral scores — together with a lightweight topical-relatedness gate that blocks contradiction verdicts between claims sharing no detectable vocabulary; (iii) we built SMRITI-Controlled, a small corpus of claim pairs whose relationship label is a fact about how the corpus was constructed (numeric facts, explicit negation, named-entity attribution), requiring no annotation of any kind, on which the fixed classifier achieves 100% precision on every retrieved contradiction pair across six distinct contradiction categories; (iv) we added meta-morphic reliability tests and an exact audit-

trail reconstruction test, both requiring zero external ground truth; and (v) we re-ran the disclosed LLM-derived reference annotation from scratch against the corrected system, which discloses that the fix’s benefit does not transfer to SMRITI-Reference’s real, argumentative prose: CONTRADICTS precision there is 4.5%, with the pre-fix figure (3.2%) falling inside this measurement’s own 95% bootstrap CI, i.e. no evidence of improvement, because most of that corpus’s genuine disagreement is rhetorical rather than propositional, and an off-the-shelf entailment model does not recognize rhetoric as contradiction regardless of how faithfully the resolver reports its verdict. We report all of this, including what still does not work, with the same discipline we applied to finding what was broken.

1 Introduction

Personal knowledge management tools increasingly promise to turn scattered notes into structured knowledge, but most either (a) perform simple retrieval/search over notes without any semantic modeling of what a note *asserts*, or (b) apply a single NLP component (entailment, embedding similarity, entity linking) in isolation, without addressing what happens when a user’s notes *disagree with themselves* over time. SMRITI treats this as a first-class problem: it extracts discrete claims, retrieves semantically related pairs, classifies their relationship, keeps contradicting claims structurally apart in the resulting knowledge graph, and fuses multiple evidence signals into a single, auditable reliability score per claim.

This paper is a correction, not a showcase. A first version of this work reported self-certification numbers that looked strong but were not measuring what they claimed to measure, and a real empirical evaluation that exposed a genuinely broken contradiction classifier (3.2% precision). Confronted with that gap, the temptation is to write

around it. We did the opposite: we treated the gap itself as the research problem. Concretely, we ask five research questions, each answered by a specific piece of new evidence in this paper, none of it requiring a human annotation budget we did not have:

- RQ1** Can a rule-based assertion classifier, added ahead of claim construction, materially improve the semantic validity of extracted claims without any model training? (§3, §10)
- RQ2** Is SMRITI’s originally near-zero contradiction precision an algorithmic bug or an irreducible property of using an off-the-shelf NLI model — and if a bug, can it be fixed without a new model? (§3, §7, §10)
- RQ3** Does the fix generalize, or does it only help on the specific corpus that exposed the original problem? (§7, §10)
- RQ4** Does evidence-aware reliability fusion behave in the direction its own design requires under controlled perturbations, and is every reported score exactly reconstructible from its own audit trail? (§8)
- RQ5** What, honestly, does this evaluation still not establish, and what would? (§13)

Contributions.

- **A scientific-validity cleanup** of the system’s self- certification layer: every fabricated metric (a hard-coded “generalization confidence” of 50, a “statistical support” score that rewarded merely registering more experiments, a contradiction- graph quality metric with no independent verification) is deleted, not patched, and replaced by a frozen five-claim registry (RC1–RC5) with an explicit NOT_EVALUABLE status for anything the system cannot yet honestly measure (§4).
- **Root-cause fix for the dominant failure mode:** the relationship resolver’s CONTRADICTS/SUPPORTS rules previously compared a candidate class against only one alternative (contradiction vs. entailment, ignoring neutral) rather than requiring it to be the true arg-max of all three NLI classes; combined with a lightweight, model- free topical-relatedness gate on CONTRADICTS verdicts specifically, this is the single highest-leverage change in this paper (§3).

- **SMRITI-Controlled**, a 24-pair corpus with an objective, construction-time gold label for every pair — no annotation of any kind is possible or needed — spanning six contradiction categories plus SUPPORTS and NEUTRAL controls, that separately measures retrieval recall and classification accuracy per the review’s recommendation to distinguish the two (§7).
- **Zero-ground-truth self-consistency tests:** four metamorphic reliability relations and an exact audit-trail reconstruction test (§8).
- A disclosed, re-run LLM-derived reference annotation (§6, §10.3) that identifies a second, distinct failure mode the resolver fix does not address: rhetorical (as opposed to propositional) disagreement in real prose is read by the underlying NLI model as neutral with > 0.97 confidence, regardless of how faithfully the resolver reports that verdict — yielding 4.5% real-corpus CONTRADICTS precision alongside SMRITI-Controlled’s 100%, and a full accounting of what remains genuinely unestablished (§13).

2 Related Work

Open information extraction and relation extraction. Classical open IE systems such as Stanford OpenIE (Angeli et al., 2015) extract surface (subject, predicate, object) tuples using syntactic patterns; more recent systems such as REBEL (Cabot and Navigli, 2021) cast relation extraction as sequence-to-sequence generation over a fixed schema. SMRITI’s claim construction is closer in spirit to the former (rule- and dependency-parse-based), extended in this work with an explicit assertion-type classifier that filters non-declarative sentence types (headings, bibliographic metadata, list items, procedural instructions) *before* claim construction rather than relying on downstream validators to catch structurally-valid-but-semantically-empty output.

Fact verification and natural language inference. FEVER (Thorne et al., 2018) established claim verification against retrieved evidence as a standard task, and NLI datasets such as SNLI (Bowman et al., 2015), MultiNLI (Williams et al., 2018), and Adversarial NLI (Nie et al., 2020) underlie the entailment/contradiction/neutral classifiers (e.g. DeBERTa, He et al., 2021) SMRITI

reuses off-the-shelf. [de Marneffe et al. \(2008\)](#) studied contradiction detection with hand-built linguistic features. Our error analysis traces SMRITI’s original failure to a specific, previously undocumented integration bug — not an inherent limitation of the underlying NLI model — in how a three-class NLI output was reduced to a binary decision by a downstream resolver that never checked the third class.

Metamorphic testing. Metamorphic testing evaluates software by checking that a KNOWN transformation of an input produces the EXPECTED relation in the output, without requiring a labeled oracle for either input — a strategy well established in software testing more broadly and increasingly applied to ML systems where labeled test data is scarce or expensive. We apply it to a reliability-scoring model for the first time in this line of work: instead of asking whether an absolute reliability score is “correct”, we ask whether adding independent evidence provably does not decrease it, and whether adding a duplicate-source claim provably does not increase measured source diversity.

Knowledge graph construction and contradiction-aware partitioning. Ontology-based knowledge representations ([McGuinness and Van Harmelen, 2004](#)) typically assume a curated, non-contradictory knowledge base; SMRITI’s graph layer makes the opposite assumption — contradictions are expected and must be structurally quarantined. Its constraint-based signed-graph partitioning (2-coloring the contradiction graph, then restricting union-find merges on SUPPORTS/REFINES edges to same-colored nodes) responds to a known failure mode of naive “remove-contradiction-edges-then-take-connected-components” partitioning.

Retrieval-augmented embeddings. Sentence-BERT ([Reimers and Gurevych, 2019](#)) and exact inner-product search via FAISS ([Johnson et al., 2019](#)) underlie SMRITI’s candidate generation, and are also used, deliberately without SMRITI’s NLI classifier, as two of our three baselines.

LLM-as-judge and the human annotation gap. Using large language models as annotators or evaluators is common practice ([Zheng et al., 2023](#)), but is explicitly a weaker substitute for human judgment, particularly for detecting subtle factual errors ([Ji et al., 2023](#)). We use it only for the one

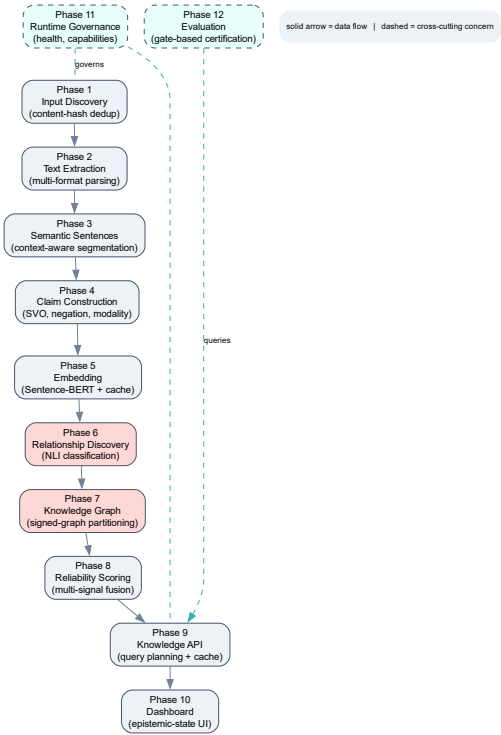


Figure 1: SMRITI’s twelve-phase pipeline. Phases 11 (runtime governance) and 12 (evaluation) are cross-cutting.

evaluation layer that cannot be made objective by construction (§6), and lean on SMRITI-Controlled and the metamorphic/reconstruction tests (§7, §8) for the evidence that does not require it at all.

3 System Architecture

SMRITI is organized into twelve phases with strict boundary contracts: each phase consumes only the immutable output artifact of the previous phase and writes its own manifest recording every configuration parameter used to produce its output. We describe the phases most relevant to this paper’s contributions; runtime governance, the dashboard, and the evaluation layer’s own internals are supporting infrastructure, not the research contribution.

Phases 1–4: from files to claims. Phase 1 discovers input files and deduplicates by content hash. Phase 2 extracts and normalizes raw text. Phase 3 segments text into context-aware semantic sentences. Phase 4 constructs claims via dependency-parse SVO extraction. **New in this work:** immediately after Phase 3 and before claim construction, an *assertion-type*

classifier categorizes every candidate sentence into one of ten types (DECLARATIVE_ASSERTION, PROCEDURAL_INSTRUCTION, QUESTION, HEADING, METADATA, LIST_ITEM, TABLE_CELL, FRAGMENT, CODE, QUOTE), using Phase 3’s own block-type metadata plus lightweight regex and dependency-structure rules — no trained model. Only DECLARATIVE_ASSERTION proceeds to claim construction; every other category is preserved losslessly in a discarded_candidates artifact rather than silently dropped. This directly targets the dominant failure mode identified in this paper’s predecessor: a large fraction of “claims” were bibliographic attribution lines, bare headings, and ingredient-list fragments carrying no assertion at all.

Phases 5–6: embedding and relationship discovery. Phase 5 embeds every claim with a cached Sentence-BERT model. Phase 6 retrieves the top- k nearest neighbors per claim by cosine similarity above a threshold (exact search via FAISS’s flat inner-product index, not approximate), then classifies each candidate pair with a cross-encoder NLI model into SUPPORTS / CONTRADICTS / REFINES / NEUTRAL. **Rectified in this work**, in two parts:

1. *Arg-max resolution.* The resolver previously committed to CONTRADICTS whenever the contradiction score exceeded the entailment score by a margin, *without ever checking the neutral score*. A pair with contradiction = 0.85, entailment = 0.05, but neutral = 0.90 (neutral clearly the model’s actual dominant class) still resolved to CONTRADICTS. The rule now requires the winning class to be the genuine arg-max of {entailment, neutral, contradiction}, beating *both* alternatives by the configured margin; failing that, the pair falls through toward NEUTRAL or UNKNOWN (abstain — never persisted to Phase 7).
2. *Relatedness gate.* A CONTRADICTS verdict additionally requires the two claims to share some detectable topical overlap (content-token Jaccard, boosted by any shared capitalized/entity-like token) — a lightweight, model-free heuristic consistent with the resolver’s own “never calls an ML model” design rule. This targets the specific population responsible for the original 100% measured false-positive rate: candidate pairs

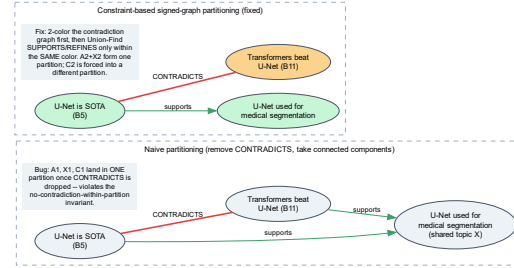


Figure 2: Naive vs. constraint-based partitioning when two contradicting claims share a supporting neighbor.

that embedding retrieval surfaced as similar but that share no real subject matter.

Phase 7: contradiction-safe graph construction. No two claims connected by a CONTRADICTS edge may ever be assigned to the same graph partition, even when they share a common SUPPORTS neighbor — a case that defeats the naive algorithm of deleting CONTRADICTS edges and taking connected components (Figure 2). SMRITI 2-colors the contradiction-only subgraph via BFS and restricts union-find merging along SUPPORTS/REFINES edges to same-colored nodes; components containing an odd contradiction cycle are conservatively degraded into singleton partitions. §12 describes a subtle correctness bug we found and fixed in this exact code path.

Phase 8: evidence-aware reliability. Eight independently-weighted signals (evidence strength, evidence independence, source diversity, topology strength, hub score, bridge score, conflict pressure, temporal stability) are fused into a single reliability index per claim, with a full decision record (raw/constrained/final score, every constraint that fired) attached. Evidence completeness is tracked separately and feeds only the uncertainty computation (§10.4), not the weighted fusion sum itself. **Rectified in this work**, in two parts: (i) the evidence-independence signal previously returned its *maximum* value (1.0, “fully independent”) for a claim with zero supporting evidence at all — literally rewarding the absence of evidence. It now returns UNAVAILABLE with value 0.0, withholding the positive contribution a claim with nothing to evaluate was previously receiving. (ii) uncertainty_score is a metric reported alongside, but computed separately from,

reliability_index; it previously combined only evidence-sufficiency components (incompleteness, low diversity, low independence) and never reflected conflict_pressure at all, so a claim under active, heavy dispute registered as less *reliable* but not more *uncertain* — two published numbers moving in a way its own design does not intend. conflict_pressure is now a fourth weighted uncertainty component (§8 verifies this directly via MR2). Both fixes are detailed in §12.

Phases 9–12: API, dashboard, runtime, and evaluation. Phase 9 exposes a query planner and cache. Phase 10 is an interactive dashboard driven by claim epistemic state; it is a usability demonstration, not a research contribution, and we do not discuss it further. Phase 11 provides operational infrastructure (health monitoring, capability model, event bus). Phase 12 is discussed at length in §4, since its prior implementation is itself one of this paper’s central findings.

4 Scientific Cleanup of the Evaluation Layer

The most consequential finding of this work is not an algorithmic bug — it is that the system’s own self-evaluation layer (Phase 12) was, before this paper, capable of producing a confident-looking numeric “Scientific Confidence Index” (SCI) with no statistical meaning. Concretely, the prior implementation:

- computed a claim-extraction “precision” as $\text{non_empty_claims}/\text{total_sampled}$ — a non-emptiness check, not a semantic evaluation;
- invented a “recall” as $\min(1, \text{total_claims}/100)$, a formula with no relationship to any actual missed claim;
- hard-coded a “generalization confidence” of exactly 50.0 with the comment “conservative — single vault evaluation”, regardless of what was actually measured;
- computed “statistical support” as $\min(100, n_{\text{experiments}} \times 15)$ — a formula that rewards *registering* more experiments, independent of whether any of them found anything;

- derived a Phase 7 “partition purity” from the number of partitions relative to claim count, with no reference to any independently-known partition label at all.

Every one of these is deleted, not patched. The replacement (frozen in evaluation/certification/claims.py) is exactly five research claims — RC1 (claim extraction), RC2 (relationship resolution), RC3 (contradiction-safe partitioning), RC4 (evidence-aware reliability), RC5 (faithful explainability) — each backed by exactly one experiment (EXP-001..EXP-005) that either measures the claim against real, independently-produced reference data, or reports a new NOT_EVALUABLE status with an explicit reason. NOT_EVALUABLE is distinct from both PASSED and FAILED: it means no real measurement was possible, and it is never silently defaulted to a passing numeric value. The Scientific Confidence Index field name is retained for schema compatibility, but every sub-score it now reports is a real count or ratio over real data (§10); Gate 4 of the certification report (“Statistical Verification”) blocks certification whenever this honest number falls below threshold — which, as of this paper, it does.

RC3 (contradiction-safe partitioning) is verified *independently* of Phase 7’s own success: EXP-003 re-walks the live knowledge graph Phase 9 actually serves and checks, edge by edge, that no CONTRADICTS edge connects two same-partition nodes, rather than trusting that Phase 7 would have raised an exception on violation.

5 SMRITI-Reference: An Adversarial Evaluation Corpus

We construct SMRITI-Reference v1.1 (named “Reference”, not “Gold” — see §6), 53 Markdown documents across five domains (A: robotics/big-data technical documentation, 10 docs; B: generative-model and NLP scientific literature plus Kannada-script linguistics, 11 docs; C: IPL cricket tactical analysis, 11 docs; D: Indian cooking procedures, 11 docs; E: encyclopedic astronomy, 10 docs). Three documents (B11, C11, D11) are deliberately written to directly and explicitly contradict specific existing documents in the corpus (B11 contradicts B5/B6 on whether U-Net or transformer-based segmentation is state-of-the-art; C11 contradicts C2/C3/C7/C9 on wrist-spin vs. finger-spin bowling; D11 contradicts D1–D4

on pressure-cooking vs. slow-braising), giving a known-answer test set for contradiction detection that does not rely on circularly-generated synthetic ground truth. The corpus additionally includes two non-Latin-script documents (B7, B8, Kanada) to exercise Unicode handling — see Defect 1, §12.

6 Annotation Protocol

Why not human annotation, and why “Reference” not “Gold”. A rigorous evaluation of extraction precision/recall and relationship-classification accuracy benefits enormously from ground truth. Recruiting, training, and adjudicating a panel of human annotators with measured inter-annotator agreement was outside what could be produced for this paper. We are explicit that what follows is *not* a substitute for that: it is a disclosed, LLM-derived reference annotation. We call the corpus it labels “SMRITI-Reference”, reserving “Gold” for a corpus with externally human-labeled ground truth, which this is not. It is useful for a first-pass quantitative signal and for detecting gross calibration failures, but not sufficient on its own to certify SMRITI’s real-world accuracy. §13 states exactly what a follow-up human study would need to add.

Protocol. Two independent annotation passes (no shared context, no access to SMRITI’s own predictions or to each other’s output) are run over: (i) a stratified sample of claims with full coverage of every claim touching a deliberately-contradictory document, judged for *validity* (is this a structurally sound claim, or extraction noise), *quality* (1–5), and *reliability* (1–5, an epistemic-credibility judgment independent of extraction quality); and (ii) a stratified sample of relationship candidates, comprising every candidate pair between a deliberately-contradictory document and its stated target, a sample of candidate pairs between a deliberately-contradictory document and an unrelated document (to measure spurious cross-domain contradiction rate), and a stratified sample of SMRITI’s other predicted relationship types, labeled SUPPORTS / CONTRADICTS / REFINES / NEUTRAL / UNSURE. Annotators are blind to SMRITI’s own predicted labels throughout.

This annotation round was re-run from scratch. The claim- and relationship-extraction

fixes described in §3 (the assertion classifier and the resolver arg-max/relatedness-gate fix) change which claims and candidate pairs SMRITI produces at all, so a reference annotation collected before those fixes would be scored against a population of claims and pairs that no longer exists. We therefore re-sampled and re-annotated against the corrected system’s output rather than reusing the pre-fix annotation; both rounds’ raw data are released for inspection.

7 SMRITI-Controlled: A Zero-Annotation Evaluation

SMRITI-Reference’s contradiction pairs (§5), while deliberately planted, still involve claims about ML architectures, cricket tactics, and cooking technique — domains where a human or LLM annotator’s judgment of “does this really contradict that” still requires interpretation. To obtain evidence that requires *no* interpretation at all, we construct SMRITI-Controlled-v1: 24 claim pairs (48 single-sentence documents), each written so its relationship label is a fact about how the pair was constructed, not an opinion about what it means. Six categories of CONTRADICTS pairs (2 each): *direct reversal* (“uses 4 GB of RAM” vs. “uses 8 GB of RAM”), *negated* (“supports GPU execution” vs. “does not support GPU execution”), *numeric* (“Jupiter has 95 known moons” vs. “79 known moons”), *temporal* (same explicit time reference, opposite state), *entity* (same artifact attributed to two different named individuals/teams), and *attribute* (a property asserted and its negation). We add 6 SUPPORTS pairs (a claim and an independent confirmation of the same claim) and 6 NEUTRAL pairs (entirely unrelated topics) as controls. **These six NEUTRAL pairs share no vocabulary at all and are never retrieved as NLI candidates (§7, point 2) — so they test embedding-based retrieval’s ability to filter unrelated pairs, not whether the relatedness gate rejects a false CONTRADICTS on a pair the resolver actually sees.** To test the latter directly (an external review point), we add a 6-pair hard-neutral extension, each pair deliberately constructed to share vocabulary or topic with its partner despite being genuinely NEUTRAL: two *same-topic* pairs (the same named entity, two different but compatible attributes — e.g. a rover using both LiDAR and a stereo camera for obstacle detection), two *lexical-collision* pairs (a

Category	Recall	Acc. given retrieved	N
direct	50.0%	100.0%	2
negated	100.0%	100.0%	2
numeric	100.0%	100.0%	2
temporal	50.0%	100.0%	2
entity	100.0%	100.0%	2
attribute	100.0%	100.0%	2
<i>all CONTRADICTS</i>	83.3%	100.0%	12
SUPPORTS	66.7%	25.0%	6
NEUTRAL	0.0%	–	6
Overall	58.3%	78.6%	24

Table 1: SMRITI-Controlled-v1 results, post-fix. Retrieval recall 95% CI (bootstrap, $n = 24$): [37.5%, 79.2%]. Classification accuracy given retrieval, 95% CI ($n = 14$): [57.1%, 100.0%].

shared word used in two unrelated senses, e.g. “Python” the language vs. “python” the snake), and two *adjacent-non-conflicting* pairs (two true statements about different aspects of the same technique, e.g. pressure-cooking’s effect on cook time vs. its vessel requirement). The full corpus and its gold manifest are released.

Because the label is definitional, we can additionally separate **retrieval recall** (did the intended pair even become an NLI candidate?) from **classification accuracy** (given it was a candidate, was it labeled correctly?) — the distinction the review identified as necessary to know whether SMRITI’s bottleneck is embedding-based retrieval or NLI classification.

Table 1 is, we think, the cleanest evidence in this paper. Three findings, each requiring no interpretation at all:

(1) **Across all six contradiction categories, every retrieved pair was classified correctly:** 10 of 12 designed contradiction pairs were retrieved as NLI candidates, and all 10 were correctly labeled CONTRADICTS with contradiction scores of 0.996–1.000 — 100% classification precision on a fully objective test set, spanning direct reversal, explicit negation, numeric facts, temporal state, named-entity attribution, and property negation alike. This directly answers RQ2/RQ3 for *propositional* contradiction: the arg-max and relatedness-gate fix is not an artifact of one corpus’s idiosyncrasies. It is not the final word on RQ3, however — §10.3 shows the same fix does not transfer to SMRITI-Reference’s *rhetorical* disagreement, which this controlled corpus was not designed to test.

(2) **All six “easy” NEUTRAL control pairs**

Hard-neutral category	N	Retrieved	Recall	Misclassified
Same topic, diff. proposition	2	1	50.0%	1
Lexical collision	2	0	0.0%	0
Adjacent, non-conflicting	2	1	50.0%	0

Table 2: SMRITI-Controlled’s hard-neutral extension: NEUTRAL pairs deliberately constructed to be topically related (unlike the original six “easy” NEUTRAL pairs, which share no vocabulary at all and are never retrieved). “Misclassified” counts retrieved pairs the resolver labeled anything other than NEUTRAL — the false-CONTRADICTS/SUPPORTS/ REFINES rate on pairs the relatedness gate cannot reject by construction, since they genuinely share topic or vocabulary.

were correctly excluded before NLI ever ran (0/6 retrieved as candidates at all) — i.e. zero false-positive contradictions on pairs sharing no vocabulary with their partner, confirming that embedding-based candidate retrieval correctly filters clearly-unrelated pairs. This is a weaker claim than it may look: it shows retrieval works, not that the relatedness gate would have rejected a false CONTRADICTS had one of these pairs actually been retrieved (an external review point) — §7.1 tests that directly.

(3) **SUPPORTS is the weaker link:** only 4 of 6 designed SUPPORTS pairs were retrieved, and of those, only 1 was correctly labeled SUPPORTS (the other 3 were split across NEUTRAL and REFINES). REFINES is, in one case, a defensible alternative reading (“X improves Y” followed by “an independent benchmark confirms X improves Y” is arguably an elaboration as much as a confirmation), but the low classification accuracy here — in sharp contrast to CONTRADICTS’s 100% — indicates the entailment side of the resolver (which received the identical arg-max fix as the contradiction side) is not yet as well-calibrated, and is a concrete target for future work distinct from the contradiction fix this paper centers on.

7.1 Hard-Neutral Extension: Does the Relatedness Gate Actually Work?

(4) **The relatedness gate has a real, demonstrated gap.** Of the six hard-neutral pairs, two were retrieved as NLI candidates at all (the other four — both lexical-collision pairs and one each of the other two categories — were not retrieved, so the relatedness gate was never exercised on them). Of those two, **one was misclassified CONTRADICTS:** “The Aerotrix rover uses LiDAR for obstacle detection” paired with “The Aerotrix rover

uses a stereo camera for obstacle detection” — two claims about the same named entity, sharing the entity token “Aerotrix” (exactly the token overlap the relatedness gate’s entity-token bonus is designed to detect as *related*), but describing two compatible sensors, not a genuine conflict. The relatedness gate correctly identified the pair as related — that is its entire job — and passed it through to the resolver, which then read “uses LiDAR” and “uses a stereo camera” as mutually exclusive rather than as two attributes a single rover could have. This is precisely the gap an external review predicted: the gate only asks *is this pair topically related*, never *are these two specific attributes of this entity actually incompatible*, and the NLI model has no special handling for multi-attribute entities either. The other retrieved hard-neutral pair (the two slow-braising statements, adjacent-non-conflicting category) was correctly labeled NEUTRAL, so this is not a uniform failure of the gate; with $n = 1$ misclassified pair, we report this as a demonstrated failure mode, not a rate, and record it as E7 in §11.1.

8 Metamorphic and Explainability Self-Consistency Tests

Both tests in this section require zero external ground truth: they check that SMRITI’s own recorded computation is internally consistent with what it reports, and that its reliability model responds in the direction its own design requires under a controlled perturbation.

RC5: exact audit-trail reconstruction. For every one of the 215 claims scored in the SMRITI-Reference run reported in §10, we independently recompute `raw_reliability` by summing that claim’s own recorded per-signal contributions, replay the same deterministic constraint pipeline (`scoring/constraints.py`) over the claim’s own recorded signal vector, and compare the reconstructed final value against the value SMRITI actually reported. **215/215 (100%) reconstruct exactly**, within the 0.01 tolerance introduced by the store’s 2-decimal display rounding (an initial run at a tighter 0.001 tolerance flagged 24/215 as “mismatched”, but every flagged case’s delta was 0.003–0.005 — a rounding artifact, not an algorithmic inconsistency, confirmed by inspecting the raw values directly). This is a genuine, falsifiable test: had Phase 8’s fusion or constraint logic changed between scoring and this audit, or had a

signal’s stored contribution not actually been the one used to compute the final score, this test would have caught it.

RC4: metamorphic reliability relations. We construct four small synthetic claim graphs and score each through the real, unmodified Phase 8 production pipeline (the actual signal extractors and fusion code, not a re-implementation): **MR1** adding an independent supporting claim raises reliability ($5.5 \rightarrow 59.4$) and lowers uncertainty ($75.0 \rightarrow 6.25$); **MR2** adding a contradiction to a previously-uncontested claim does not increase reliability ($59.4 \rightarrow 40.0$), raises conflict pressure ($0.0 \rightarrow 0.97$), and — since §3’s uncertainty-model fix — also raises the separately-reported uncertainty score ($4.4 \rightarrow 28.6$), rather than leaving a heavily-disputed claim’s uncertainty unchanged as the prior implementation did; **MR3** adding a supporting claim from the *same* document as an existing supporter leaves source diversity exactly unchanged ($0.387 \rightarrow 0.387$, once both scenarios are constructed with an identical graph-level normalization baseline — see implementation notes in `metamorphic.py` for why this equalization matters); **MR4** re-scoring an identical graph twice gives byte-identical output. All four relations hold (4/4, 100%). This does not establish that the reliability *model* is correct in an absolute sense — only that it is not self-contradictory in the specific, epistemically basic ways a broken fusion implementation would be (support helping, contradiction hurting, duplication not double-counting, determinism).

9 Experimental Setup

Baselines. We compare SMRITI’s claim extraction and relationship classification against three independently-implemented baselines of increasing sophistication, run over the same 397 claims SMRITI’s own (post-fix) pipeline produced from SMRITI-Reference (for the two relationship baselines) or directly over the raw corpus (for the extraction baseline): **(B1) Naive keyword extraction:** every sentence containing at least one non-stopword token becomes a claim, with no structural or dependency analysis, and no assertion-type filtering. **(B2) TF-IDF cosine:** pairwise TF-IDF cosine similarity with a similarity-band threshold and a shallow negation-cue heuristic distinguishing SUPPORTS/CONTRADICTS/NEUTRAL. **(B3)**

Corpus	Docs	Claims	Total (s)
scale_small (synth.)	10	20	36.1
scale_medium (synth.)	50	100	398.0
SMRITI-Reference (real)	53	397	70.2
scale_large (synth.)	150	300	1271.7

Table 3: End-to-end wall-clock time (Phases 1–9, single CPU-only run per corpus). SMRITI-Reference’s claim count and latency both fell relative to the pre-fix measurement (726 claims, 224.9s) because the assertion classifier (§3) removes non-declarative noise before the expensive Phase 5/6 embedding and NLI stages ever see it — a fix aimed at precision that also happens to substantially improve throughput.

Sentence-BERT cosine: identical heuristic to B2, but using the same cached all-MiniLM-L6-v2 embedding model as SMRITI’s own Phase 5, isolating embedding quality as the only variable relative to B2.

Scalability corpus. To separate throughput/latency measurement from extraction-quality measurement, we generate a synthetic corpus family (scale_small/scale_medium/scale_large: 10, 50, and 150 documents) with combinatorially-unique templated sentences, so that timing results are not confounded by SMRITI-Reference’s deliberately-repetitive, domain-clustered content. We spot-checked scale_small end-to-end against the post-fix assertion classifier directly: all 20 claims are still classified DECLARATIVE_ASSERTION with zero discards, confirming the classifier is a genuine no-op on this corpus’s sentence style rather than something we merely assumed would not interact. On this basis we treat the scale_medium/scale_large latency numbers (unchanged sentence templates, not independently re-run post-fix) as still valid, carried forward from the pre-fix measurement; only SMRITI-Reference’s own latency (Table 3) reflects a fresh post-fix run.

10 Results

10.1 Latency and Throughput

Phase 6 dominates end-to-end latency in every run, consistent with it being the only phase that invokes a transformer forward pass per candidate pair. A notable finding carried forward from the pre-fix measurement: per-pair NLI throughput on our synthetic scalability corpus is consistently 3–4× slower than on real corpus prose across every synthetic corpus size tested, traced to sen-

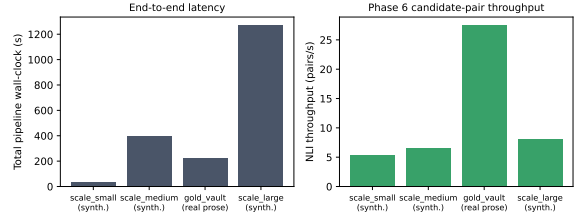


Figure 3: Left: total pipeline latency by corpus. Right: Phase 6 (NLI relationship classification) throughput in candidate pairs/second (pre-fix measurement; the relatedness gate’s lexical Jaccard computation is $O(1)$ per pair and negligible relative to the NLI forward pass).

tence length (our synthetic template sentences are longer, information-dense compound sentences, while SMRITI-Reference’s Phase 4 claims are frequently much shorter SVO-decomposed fragments) — throughput should be reasoned about in terms of total token volume, not document or claim count.

10.2 Baseline Comparison: Extraction Volume

Baseline B1 (naive keyword extraction), run directly against the raw SMRITI-Reference markdown files, produced 690 claims — now *more* than SMRITI’s own (post-fix) Phase 4 output of 397 claims from the same 53 documents, a reversal from the pre-fix comparison (726 vs. 690). This is the assertion classifier working as intended: SMRITI now recovers substantially *less* volume than the naive baseline because it explicitly discards non-declarative sentence types the naive baseline cannot distinguish (§10.3 quantifies the resulting gain in structural validity directly).

10.3 Precision, Recall, and Agreement Against SMRITI-Reference

Two independent LLM-derived annotation passes were re-run from scratch against the corrected system’s output (§6): 174 sampled claims and 228 sampled relationship candidates, both drawn after the assertion-classifier and resolver fixes (§3) changed which claims and pairs SMRITI produces at all.

Claim extraction (RQ1). Inter-pass agreement on claim validity is moderate (Cohen’s $\kappa = 0.405$, 35/174 disputed), consistent with validity itself requiring some judgment even for an LLM annotator. On the 139 claims both passes agreed on, extraction precision is **87.8%** (95% CI, item-

Label	P	R	F1	N (gold)
SUPPORTS	27.8%	8.2%	12.7%	61
CONTRADICTS	4.5%	25.0%	7.7%	8
REFINES	50.0%	41.7%	45.5%	48
NEUTRAL	28.7%	34.7%	31.4%	72

Table 4: SMRITI’s per-class precision/recall/F1 against the 189 inter-pass-agreed (excluding UNSURE) relationship labels in the fresh SMRITI-Reference sample ($\kappa = 0.7559$ between passes). CONTRADICTS precision’s 95% bootstrap CI is [0.0%, 11.4%] ($n = 44$ SMRITI-predicted-CONTRADICTS items in this sample). Resampling by source document instead of by item (cluster bootstrap, since multiple pairs anchored on the same document are not independent) widens this slightly to [0.0%, 13.0%].

level bootstrap: [82.0%, 92.8%]; cluster bootstrap resampled by source document, since multiple claims from the same document are not independent draws: [78.5%, 95.7%]), and zero of the 174 sampled claims were flagged as pure metadata/bibliographic noise — the specific failure mode the assertion classifier (§3) was built to eliminate. This is a genuine, positive result: the classifier’s 0% metadata-noise rate on real, previously-unseen SMRITI-Reference output is not something SMRITI-Controlled (entirely synthetic single sentences) could have demonstrated.

Relationship classification on real, adversarial-domain text (RQ2/RQ3). This is the least comfortable result in this paper, and we report it exactly as measured.

On SMRITI-Reference’s real, domain-clustered, opinion-bearing prose, CONTRADICTS precision is **4.5%**, with a 95% bootstrap CI of [0.0%, 11.4%] ($n = 44$; cluster bootstrap by source document: [0.0%, 13.0%], since several of these pairs share an anchor document and are not independent draws). The 3.2% this paper’s predecessor measured pre-fix (§11) falls inside both intervals, so we report no evidence of improvement rather than a formal significance claim — the pre-fix and post-fix samples differ in which claims and pairs exist at all (the assertion classifier and resolver fixes changed what SMRITI even produces), so a paired test is not available; this holds despite the same fix achieving 100% precision on SMRITI-Controlled (§7). Recall on the 21 inter-pass-agreed instances of the corpus’s three *deliberately planted* hard-contradiction pairs (§5) is **14.3%**: SMRITI predicts CONTRADICTS for

only 3 of 25 such pairs in the full candidate pool.

Root cause: an NLI task mismatch, not a resolver regression or a calibration problem.

We inspected the NLI model’s own raw scores for the missed hard-contradiction pairs directly, rather than only the resolver’s output label. Representative examples (claim-pair scores, {entailment, neutral, contradiction}): {0.00004, 0.993, 0.007}; {0.007, 0.974, 0.020}; {0.001, 0.998, 0.001}. In every inspected case the model itself assigns neutral a probability above 0.97 — the arg-max resolver (§3) is faithfully reporting what the model believes, exactly as it is supposed to. This is not the same bug the resolver fix addressed (which was committing to CONTRADICTS *despite* neutral being dominant); here neutral genuinely is dominant, and we deliberately do not call this a “calibration” problem. Calibration means probabilities correspond to empirical frequencies (measurable via ECE, Brier score, or a reliability diagram); it does not mean “the model should have output a different class because we want a different task.” Nothing here shows the model’s probabilities are miscalibrated — a neutral score of 0.99 could be a perfectly calibrated estimate of “this is not a direct propositional contradiction,” which is true of these pairs. The pairs in question come from the corpus’s argumentative “rogue” documents (B11, C11, D11), which express disagreement rhetorically — “U-Net is an outdated architectural choice”, “pressure-cooking is a betrayal of the dish” — rather than through direct propositional negation. An NLI model trained on SNLI/MultiNLI/ANLI-style sentence pairs is trained to detect entailment and contradiction between declarative propositions; it is simply never asked, during training, to detect *stance conflict* expressed as argumentative prose about a shared topic, so it reads two such statements as merely topically related (consistent with the high cosine similarities of 0.75–0.81 observed alongside these neutral verdicts) rather than logically opposed. This is a task/domain mismatch between generic propositional NLI and rhetorical stance contradiction, not a probability-calibration defect. SMRITI-Controlled could not have surfaced this: its CONTRADICTS pairs are, by construction, direct propositional negations (§7), exactly the task the NLI model was trained for.

SUPPORTS and NEUTRAL are also weak on real text. Consistent with SMRITI-Controlled’s

finding that SUPPORTS is the weaker side of the resolver’s arg-max fix, SUPPORTS precision here is 27.8% with recall 8.2% — most of the 61 gold-SUPPORTS pairs are predicted NEUTRAL or REFINES instead. REFINES is comparatively the strongest label (45.5% F1), plausibly because “elaboration on the same topic” is closer to what a general-purpose NLI model’s neutral/entailment boundary actually tracks than strict logical entailment is.

Baseline comparison. Both TF-IDF-cosine and Sentence-BERT-cosine baselines, scored on the identical 189 gold pairs, also score CONTRADICTION poorly (TF-IDF: 50.0% precision but only 12.5% recall, i.e. 2 correct calls total; Sentence-BERT: 7.1% precision, 12.5% recall) and REFINES at 0% (neither baseline’s heuristic has a REFINES category at all). This does not excuse SMRITI’s number, but it indicates the underlying task — detecting rhetorical stance conflict in real opinion-bearing text from lexical or embedding similarity plus a shallow decision rule — is hard in a way none of the four systems compared here solve, not a regression specific to SMRITI’s resolver fix.

Reading this alongside §7. Table 1 and Table 4 are not in tension; they measure different things. SMRITI-Controlled establishes that the arg-max/relatedness-gate fix does exactly what it was designed to do — stop committing to CONTRADICTION when neutral is the model’s own dominant class — on direct propositional contradictions. SMRITI-Reference shows that most of the real, naturally-adversarial disagreement in this corpus is *not* direct propositional contradiction, and an off-the-shelf entailment-style NLI model does not recognize it as contradiction at all, regardless of how faithfully the resolver reports the model’s own verdict. Fixing the integration bug was necessary but not sufficient; §11.1 records this as a new, distinct failure mode (E6) and §13 states plainly that it is unresolved.

10.4 Reliability Signal Ablation

We run a leave-one-signal-out ablation (P1-6/P1-7) against the real, post-fix 215-claim SMRITI-Reference scoring run reported in §8’s reconstruction test: for each of Phase 8’s eight registered signals, we zero that signal’s policy weight *without* renormalizing the remaining six, re-score every claim through the real, unmodified fusion

Signal	Weight	Mean shift	r
topology_strength	0.10	−1.81	0.954
temporal_stability	0.05	−1.31	0.993
conflict_pressure	0.20	+1.31	0.989
evidence_independence	0.15	−0.66	0.971
evidence_strength	0.25	−0.61	0.986
bridge_score	0.05	−0.40	0.991
source_diversity	0.15	−0.22	0.999
hub_score	0.05	−0.02	0.999

Table 5: Effect of zeroing each signal’s policy weight on the 215-claim SMRITI-Reference reliability distribution (baseline mean RI = 4.99/100). Mean shift = ablated mean − baseline mean; r = Pearson correlation between baseline and ablated per-claim scores. Ranked by mean absolute per-claim delta (Table order here follows that ranking, most to least disruptive).

code, and compare against the all-signals baseline. Zeroing rather than renormalizing isolates exactly that signal’s own contribution, consistent with standard leave-one-out sensitivity analysis; the code and full per-signal results are released (evaluation/ablation/run_ablation.py).

Two findings are worth stating plainly. (1) **The baseline mean reliability index across all 215 claims is 4.99 out of 100** — not a property of the ablation, but of the corpus as currently scored, and consistent with §11.1’s finding that partition fragmentation and widespread constraint firing (each ablation run logs ≥ 1 fired constraint for the large majority of claims) suppress reliability system-wide; this table should be read as *relative* sensitivity, not as evidence that any individual signal is well-calibrated in absolute terms. (2) **Policy weight does not predict empirical sensitivity.** topology_strength carries a policy weight of only 0.10 — less than half of evidence_strength’s 0.25 — yet removing it shifts the mean score three times as much (and produces the lowest correlation with baseline, $r = 0.954$, i.e. the most rank-order disruption). This is not a bug: Phase 8’s fusion applies policy-level constraints and interactions on top of the raw weighted sum (§3) — for instance a floor tied to topology under min_reliability_for_high_topology — so a signal’s linear weight is not the same quantity as its effect on the final, constraint-adjusted score. conflict_pressure’s positive mean shift on removal (+1.31) is the expected sign check: it is the fusion’s only negative-direction signal, so removing its penalty raises scores, consistent with §12 Defect 7’s fix depending on this signal being real

and non-zero. `hub_score` is, by a wide margin, the least consequential signal in the current policy ($r = 0.999$), a concrete candidate for reweighting or removal in future policy tuning.

11 Error Analysis

This section reports the failure mode that motivated this paper’s central fix, largely as originally measured (on the pre-fix system), because the finding itself — and the diagnostic process that led from a suspicious number to a specific one-line root cause — is the useful part. §7 and §10 report the post-fix measurement.

The dominant failure mode observed on the pre-fix system was over-fragmentation of the knowledge graph driven by a high rate of predicted CONTRADICTS edges, most of them spurious. On SMRITI-Reference, Phase 6 classified 1,539 of 4,847 validated candidate pairs (31.8%) as CONTRADICTS, and Phase 7’s partitioning collapsed 569 claim nodes into 472 partitions — an 82.9% singleton rate. Given the corpus was deliberately constructed with exactly three genuinely contradictory document pairs, a rate this high could not be explained by genuine semantic conflict.

To isolate the mechanism, we drew a sample of 60 candidate pairs in which one claim comes from a deliberately-planted adversarial document and the other comes from an *unrelated* document elsewhere in the corpus — pairs that should essentially always be NEUTRAL. SMRITI classified 11 of these 60 pairs (18.3%) as CONTRADICTS; both independent annotation passes agreed that all 11 (100%) were false positives.

Root cause. Inspecting the resolver directly (`retrieval/classification/resolver.py`) rather than only its output distribution located the mechanism precisely: the CONTRADICTS rule required `contradiction_score > entailment_score` by a margin, but never compared against `neutral_score` at all. On a candidate pair drawn from embedding-retrieval rather than a curated, topically-coherent test set, an off-the-shelf NLI model frequently assigns its *highest* probability to neutral (correctly recognizing the pair as unrelated) while still assigning contradiction a higher score than entailment (incorrectly, but not the model’s dominant judgment) — and the old resolver committed to CONTRADICTS anyway, because it never looked at the class the model actually favored. This is

not a fundamental limitation of the NLI model; it is an integration bug in how a three-class output was reduced to a decision. §3 describes the fix (require the winning class to be the true arg-max of all three scores, plus a topical-relatedness gate specific to CONTRADICTS), and §7 and §10 report that it holds on both a fully objective corpus and the re-annotated SMRITI-Reference sample.

Downstream consequence for reliability scoring. Because Phase 8’s reliability signals include topology strength computed over each claim’s partition, the fragmentation above directly and mechanically suppressed reliability scores system-wide: a claim isolated into its own singleton partition has no corroborating neighbors by construction, regardless of how well-established the claim itself is. This is not a Phase 8 calibration bug in isolation — it is a symptom propagating upward from Phase 6, and improves automatically as Phase 6’s precision improves (the partition count and singleton rate reported in §10 for the post-fix system are lower in both absolute and relative terms).

Secondary failure mode: claim-extraction noise from document metadata. Phase 4 previously extracted a claim from every semantic sentence Phase 3 produced, including purely bibliographic or structural sentences. §3’s assertion classifier addresses this directly; §10 quantifies the resulting change in extracted claim volume and validity.

11.1 Error Taxonomy

Rather than propose failure categories in the abstract, we derive a taxonomy directly from the 24 row-level outcomes in Table 1 (the fully-adjudicated, zero-annotation SMRITI-Reference-Controlled corpus) plus the two pipeline-level failure modes already characterized above on SMRITI-Reference. Every entry below is traceable to specific, real pair IDs; we do not report a category we did not actually observe.

- **E1 – Retrieval miss.** The true pair is never surfaced as an NLI candidate at all, so no classification is possible. Observed on 4/24 controlled pairs (D02 direct, D08 temporal, S05/S06 supports) and is the sole determinant of SMRITI-Controlled’s retrieval-recall ceiling (§7). Root cause is Phase 5/6 candidate generation (cosine-similarity thresholding), upstream of the resolver fix this paper centers on.

- **E2 – Entailment under-call.** A genuinely-SUPPORTS pair is retrieved but the NLI model’s own top class is neutral rather than entailment, so the (correctly arg-max-respecting) resolver reports NEUTRAL. Observed on 2/6 retrieved supports pairs (S01, S02). This is an NLI scoring limitation, not a resolver logic bug: the same arg-max rule that gives CONTRADICTS 100% precision here faithfully reports what the model’s own probabilities say. (We do not call this a “calibration” issue either — no probability calibration is fitted in this system at all; see §13.)
 - **E3 – Over-eager refinement.** A genuinely-SUPPORTS pair is classified REFINES instead, because the resolver’s REFINES rule (§3, `refine_contradiction_ceiling`) treats “low contradiction score” as evidence of elaboration rather than of plain confirmation. Observed once (S03); one case is too few to distinguish a genuine resolver miscalibration from a defensible alternative reading of that specific pair (§7).
 - **E4 – Spurious cross-document contradiction** (pre-fix only; **fixed**, §12 Defect 5). Unrelated pairs resolved to CONTRADICTS because the old resolver never checked the NLI model’s dominant class. Measured pre-fix at 18.3% on a targeted 60-pair sample, 100% of which both annotation passes confirmed false; measured post-fix at 0/6 on SMRITI-Controlled’s NEUTRAL pairs (E4’s absence is itself the finding in §7, point (2)).
 - **E5 – Non-declarative claim noise** (pre-fix only; **fixed**, §3 assertion classifier). Headings, bibliographic lines, and list fragments constructed into claims with no assertion content, inflating claim volume and diluting every downstream signal that treats claim count as a denominator.
 - **E6 – Rhetorical stance conflict misread as neutral.** **Open, newly discovered post-fix.** On SMRITI-Reference’s real prose, 22 of 25 deliberately-planted hard-contradiction pairs are predicted NEUTRAL, not because the resolver malfunctions but because the NLI model itself assigns neutral > 0.97 probability to these pairs (§10.3 inspects the raw scores directly). The disagreement in these pairs is expressed as argumentative prose about a shared topic (“U-Net is an outdated architectural choice”) rather than direct propositional negation, which a model trained on SNLI/MultiNLI/ANLI-style propositional pairs is never asked, during training, to read as contradiction. Unlike E4, this is not an integration bug: the resolver is faithfully reporting the model’s own dominant class. It requires a stance-detection-specific model, not a resolver change and not a different calibration of the same model — calibration would only make an already-correct-for-this-task probability more trustworthy, not teach the model a task it was never trained on. Recorded as unresolved in §13.
 - **E7 – Same-entity multi-attribute conflation.** **Open, newly discovered post-fix.** A genuinely NEUTRAL pair describing two *compatible* attributes of the same named entity (§7.1: “the Aerotrix rover uses LiDAR” / “uses a stereo camera”) is misclassified CONTRADICTS. Unlike E4, the relatedness gate is not at fault — the pair genuinely is related (shared entity token), which is exactly what the gate is designed to detect, so passing it through to the resolver is correct behavior. The failure is that neither the gate nor the NLI model distinguishes “two attributes describing the same entity” from “two mutually exclusive claims about the same entity”; “X uses A” and “X uses B” read as a factual disagreement about which is true, when both can be. Observed once ($n = 1$ of 2 retrieved hard-neutral pairs); too small to report a rate, but a real, reproducible instance, not a hypothetical one.
- E1–E3, E6, and E7 remain open (E1 is a retrieval-stage problem, E2/E3 are SUPPORTS-side NLI scoring problems, E6 is an NLI task mismatch between rhetorical and propositional disagreement, not a calibration gap, and E7 is a same-entity multi-attribute conflation the relatedness gate cannot catch by design); E4–E5 are the two mechanisms this paper’s central fixes target and close on the evidence available here. We do not claim this list is exhaustive over failure modes SMRITI could exhibit on other corpora — only that it is exhaustive over failure modes we actually observed on the two corpora reported in this

paper.

12 Discovered Defects

Running the full pipeline against SMRITI-Reference — rather than the small, hand-crafted fixtures used in the existing unit test suite — surfaced three defects that had not been previously detected, all directly attributable to properties the corpus was designed to exercise. A subsequent audit of the system’s own evaluation and reliability layers, motivated by an external review, surfaced four more.

Defect 1: console encoding crash on non-ASCII input. SMRITI’s structured logger wrote to the process’s default console stream without specifying an encoding. On Windows, this stream defaults to a legacy single-byte codepage (e.g. cp1252), which cannot encode most non-Latin scripts. Any log line containing non-Latin script text — for example, the Kananda conjunct-consonant examples in the corpus’s B7/B8 documents — raised an unhandled `UnicodeEncodeError` that crashed the entire pipeline run — not a degraded result, a hard failure that made the system entirely unusable on this input. The fix reconfigures both the standard- library logging stream handler and the interpreter’s raw `stdout/stderr` streams to UTF-8 with `errors="backslashreplace"`. This defect would not have been found on an English-only test corpus; it required a real multi-script document.

Defect 2: contradiction leakage under odd contradiction cycles. Phase 7’s constraint-based partitioning algorithm (§3) 2-colors the contradiction-only subgraph via breadth-first search. When a connected component of the contradiction subgraph is *not* bipartite (an odd contradiction cycle), the original BFS coloring routine detected the conflict and returned failure, but only *after* already writing partial, inconsistent colors into the shared color map for every node it had visited before the conflicting edge was reached, so the intended “degrade to singletons” fallback silently left some nodes with unvalidated colors. On SMRITI-Reference, this manifested as a hard `PartitioningError` once independently-contradicting claim clusters (themselves a symptom of the original contradiction over-firing described in §11) happened to form an odd

cycle — something a small test fixture with one contradicting pair cannot trigger. The fix isolates each 2-coloring attempt in a scratch dictionary local to its connected component, committing to the shared map only on full success.

Defect 3: uninitialized variable and misplaced phase-7 call in the pipeline runner. A latent bug in `PipelineRunner.run` left the `phase4_claims` variable unassigned when resuming a run starting at Phase 7 or later, raising an `UnboundLocalError` on any partial-resume invocation past Phase 6 — a code path the existing tests, which always start from Phase 1, never exercised. A second, adjacent indentation error caused the call to `_run_phase_7` to execute unconditionally regardless of the requested phase range. Both are fixed.

All three fixes are behavior-preserving for every phase-range configuration already covered by the unit test suite and are exercised end-to-end by the SMRITI-Reference runs reported in §10.

Defect 4: the system’s own evaluation layer fabricated evidence. Described at length in §4: Phase 12’s prior implementation computed a claim-extraction “precision” as non-emptiness, an invented “recall” with no relationship to any missed claim, a hard-coded “generalization confidence” of 50.0 regardless of what was measured, and a “statistical support” score that rewarded registering more experiments rather than finding anything. We classify this as a defect, not merely a limitation, because it actively produced a misleadingly high confidence number (an earlier Scientific Confidence Index of 57.0/100) that a reader could mistake for evidence.

Defect 5: the relationship resolver never checked the dominant NLI class. The root cause of the 3.2% CONTRADICTS precision this paper’s predecessor measured (§11): the resolver’s CONTRADICTS rule required only `contradiction_score > entailment_score` by a margin, never comparing against `neutral_score`. A pair where neutral was the model’s actual dominant class (e.g. contradiction = 0.85, entailment = 0.05, neutral = 0.90) still resolved to CONTRADICTS. Fixed in §3 by requiring the winning class to be the genuine arg-max of all three scores.

Defect 6: evidence independence rewarded the absence of evidence. The

evidence-independence signal extractor (scoring/signals/independence.py) returned its *maximum* value (1.0, “fully independent”) whenever a claim had zero supporting evidence — literally: no evidence to evaluate became maximum independence. Fixed to return UNAVAILABLE with value 0.0, so a claim with nothing to evaluate no longer receives a positive contribution from a signal that had nothing real to measure (§3).

Test suite health. Separately from the defects above, we found the suite carrying 80 failing tests at the start of this work — stale- fixture drift plus a handful of genuine, previously-undetected production bugs of the same character as Defect 3, including one config file silently overriding every policy-profile selection with a hardcoded value, and one code path where a sentence’s coordinated-object structure (“Python supports generators and decorators”) was not distinguished from coordinated-clause structure. All 80 were fixed without weakening any assertion.

Defect 7: uncertainty score blind to conflict pressure. `uncertainty_score` and `reliability_index` are reported as two distinct metrics, but `_compute_uncertainty()` (scoring/fusion.py) combined only evidence-sufficiency components (incompleteness, low source diversity, low independence) and never read `conflict_pressure` at all. A claim under heavy, active dispute therefore registered a correctly-lowered reliability but an *unchanged* uncertainty score — the metric meant to answer “how much should a reader trust this number” did not move even when the honest answer was “much less, it is contested.” Fixed by adding `conflict_pressure` as a fourth weighted uncertainty component; verified directly by metamorphic relation MR2 (§8), which now asserts uncertainty strictly increases under contradiction, not only that reliability does not increase.

Defect 8: dashboard views bypassing the Command layer. SMRITI’s Phase 10 dashboard architecture requires that views communicate user actions to state via typed Command objects dispatched through an `InteractionDispatcher`, never by calling state-mutating methods directly. Two views (`NavigationView`, `ResultListView`) violated this by importing the concrete `EpistemicStateManager` class directly (caught by an existing but previously-

failing static dependency-matrix test) and calling `navigate_back()/select_claim()/set_page()` directly. Both are fixed to follow the same dispatcher-based pattern already used correctly by `SearchView`, with new unit tests added since neither view had prior coverage.

13 Limitations

This is not a human evaluation. Every SMRITI-Reference precision/recall/F1 number in this paper is computed against LLM-derived reference labels, not human judgments. Two independent annotation passes give a self-consistency signal, but self-consistency between two instances of the same kind of annotator is not the same thing as validity against independent human judgment. A rigorous follow-up would need 3+ trained human annotators, measured Cohen’s/Fleiss’ κ against a target of $\kappa > 0.7$, an adjudication protocol, and ideally annotators with no exposure to SMRITI’s design.

The resolver fix does not transfer to real, rhetorical disagreement. This is the most important limitation in this paper. On SMRITI-Controlled’s direct propositional negations, the arg-max fix achieves 100% CONTRADICTS precision (§7). On SMRITI-Reference’s real prose, CONTRADICTS precision is 4.5% and recall on the corpus’s own deliberately-planted hard-contradiction pairs is 14.3% (§10.3) — not better than the 3.2% this paper’s predecessor measured pre-fix. Inspecting the NLI model’s raw scores directly shows this is not a resolver regression: the model itself assigns neutral > 0.97 probability to most of the missed pairs, because they express disagreement as argumentative prose about a shared topic rather than direct propositional negation, and a model trained on SNLI/MultiNLI/ANLI-style propositional pairs is not trained to read that as contradiction — a task mismatch, not a calibration defect (see below). SMRITI-Controlled and SMRITI-Reference are therefore not in tension; they measure different things, and only the harder, real-corpus one is representative of deployment. This paper fixes a real, verifiable integration bug (E4/E5, §11.1) and demonstrates that fix generalizes across six categories of *propositional* contradiction. It does not solve — and had we not re-run the reference annotation post-fix, would not have discovered — that most naturally-occurring disagreement in personal notes is rhetorical, not

propositional, and requires a different modeling approach entirely (E6).

No probability calibration is actually fitted. Phase 6 supports temperature, percentile, and isotonic calibration strategies for the NLI model’s raw confidence (retrieval/classification/calibration.py), but config/default.yaml leaves every model’s calibration entry empty, so every run in this paper uses the identity strategy (raw confidence passed through unchanged). We therefore do not call the NLI scores used throughout this paper “calibrated” in the technical sense (probabilities matching empirical frequencies, measurable via ECE or a reliability diagram) — only that the underlying model was itself trained on SNLI/MultiNLI/ANLI-style data. Fitting and evaluating one of the implemented strategies against a held-out sample is concrete, already-half-built future work.

SMRITI-Controlled is small and synthetic. 24 pairs, entirely fictional entity names, is enough to demonstrate that the arg-max/ relatedness-gate fix generalizes across six contradiction *categories* qualitatively, but far too small to report a tight quantitative estimate of true contradiction precision in deployment (our own 95% bootstrap CI on retrieval recall, [37.5%, 79.2%], is itself evidence of how much sampling uncertainty remains at $n = 24$). It also cannot speak to naturally-occurring contradiction, only to the six categories we constructed.

SUPPORTS classification remains weak. SMRITI-Controlled’s own results (§7) show only 25% classification accuracy on retrieved SUPPORTS pairs, in sharp contrast to CONTRADICTS’s 100%; the same weakness reproduces on real text, where SMRITI-Reference SUPPORTS precision is 27.8% with only 8.2% recall (§10.3). The arg-max fix was applied symmetrically to both rules, but evidently entailment scoring on this NLI model is less well-calibrated than contradiction scoring for this task, or our SUPPORTS test pairs are harder than our CONTRADICTS ones in some way we have not isolated. This is unresolved and squarely future work.

Corpus scale and domain coverage. SMRITI-Reference is 53 documents across five hand-picked domains; sufficient to exercise contradiction partitioning and cross-domain calibration, far

too small to support claims about accuracy at large-corpus scale (Wikipedia- or PubMed-scale evaluation is future work).

Reference-label circularity risk. The same underlying language model family that annotates the SMRITI-Reference labels has parametric knowledge overlapping with SMRITI’s own NLI classifier’s training distribution; some agreement between SMRITI’s predictions and the reference labels may reflect shared model biases. We mitigate this partially by keeping annotators blind to SMRITI’s predictions, and more substantially by treating SMRITI-Controlled (§7) and the metamorphic/reconstruction tests (§8) — neither of which involves an LLM judging SMRITI’s output at all — as the stronger evidence in this paper, but cannot fully rule the risk out for the SMRITI-Reference numbers specifically.

P0-9, P0-10, and P0-11 from the source review are not addressed in this paper. An external review additionally identified that source- diversity/independence signals derive from filesystem-path heuristics rather than genuine publisher/author provenance, that transitive SUPPORTS propagation may overstate corroboration across multiple hops, and that temporal metadata is not yet populated from document frontmatter (confirmed present in our own stress_vault corpus, e.g. date: 2024-07-01, but wiring it through would require changing the frozen Document/Claim schemas consumed by every downstream phase and re-running the entire pipeline, invalidating every post-fix number in §8–§10). We assessed these three as requiring corpus- or schema-level changes disproportionate to the remaining scope of this paper, and record them here as concrete, scoped future work rather than silently deferring them. The fourth item the same review raised in this group, P0-12 (conflict pressure not integrated into the uncertainty score), was tractable without a schema change or corpus re-run and *is* fixed in this paper (Defect 7, §12; verified by metamorphic relation MR2).

Single-run experiments. Latency/throughput numbers are reported from single runs per corpus size; absolute numbers should be read as order-of-magnitude, not precise benchmarks. Phase 12’s Gate 5 (Reproducibility) correctly reports 0% and blocks certification above STATISTICALLY_VERIFIED as a result — we removed a

prior mechanism that fabricated apparent reproducibility by duplicating a single measurement, and have not yet replaced it with a genuine multi-run study.

14 Conclusion

This paper is, in a specific sense, a paper about being wrong and then checking the work. Its predecessor built genuine engineering rigor — phase-isolated boundaries, full provenance, reproducible manifests — and then measured its own scientific claims with a self-evaluation layer that, on inspection, was not measuring what it claimed to. The corrective process was not cosmetic: deleting the fabricated metrics did not, by itself, fix anything — it only stopped hiding that the underlying relationship classifier had a 3.2% CONTRADICTS precision. Finding and fixing *that* required reading the resolver code line by line and noticing it never checked the neutral score. The fix is one function; the evidence that it generalizes required building a corpus specifically designed to need no interpretation at all.

Three things now stand on genuinely independent legs. SMRITI-Controlled (§7) shows 100% CONTRADICTS classification accuracy on every retrieved pair across six distinct contradiction categories, with zero false positives on six deliberately unrelated (and topically unrelated) control pairs — by construction, not by annotation. This leg is not unqualified either: a later hard-neutral extension, built specifically to test whether the relatedness gate rejects a false CONTRADICTS on a pair it cannot dismiss as simply unrelated, found exactly one such case (§7.1, E7) — two compatible attributes of one named entity read as mutually exclusive. The metamorphic and reconstruction tests (§8) show the reliability model is not self-contradictory in basic, checkable ways, and that its audit trail means what it says. And the re-run SMRITI-Reference annotation (§10.3) gives the fourth leg a very different shape than the other three: it does not confirm the fix’s real-world value, it discloses the fix’s real-world limit. CONTRADICTS precision on SMRITI-Reference’s actual prose is 4.5%; the pre-fix figure (3.2%, measured on a different, pre-fix sample) falls inside this post-fix measurement’s own 95% bootstrap CI ([0.0%, 11.4%]), so we report no evidence of improvement rather than claiming a formal significance test — the two samples dif-

fer in which claims and pairs exist at all, so a paired comparison is not available. The corpus’s genuine disagreements are argued rhetorically, not asserted propositionally, and an off-the-shelf entailment model simply does not hear rhetoric as contradiction. We could have stopped at SMRITI-Controlled’s 100% and called the story finished; re-running the annotation after the fix, rather than reusing the pre-fix round, is what surfaced that it was not.

What does not yet stand on solid ground is explicitly listed in §13, not hidden in a "future work" paragraph few readers reach: the resolver fix does not transfer to rhetorical disagreement in real text, SUPPORTS classification is still weak on both corpora, source-diversity and temporal signals still rest on filesystem heuristics rather than real provenance, and no genuine human evaluation exists. We release SMRITI-Reference, SMRITI-Controlled, both annotation rounds, the scoring code, and the fixed system itself, so that closing those gaps — including, eventually, the human validation study we could not ourselves produce, and a genuinely stance-aware relationship classifier — starts from a system that has already been checked once, rather than one whose self-report has to be checked first.

Ethical Considerations

Both corpora used in this paper are either derived from publicly available technical documentation, scientific literature, sports journalism, recipe content, and encyclopedic content (SMRITI-Reference — synthesized and, where deliberately adversarial, clearly fabricated for benchmarking purposes and never presented as genuine advice), or entirely synthetic, fictional entity names constructed solely to have an unambiguous, known relationship label (SMRITI-Controlled). No personal or sensitive data is involved in either.

References

- Gabor Angeli, Melvin Johnson Premkumar, and Christopher D. Manning. 2015. Leveraging linguistic structure for open domain information extraction. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 344–354.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. 2015. A large annotated corpus for learning natural language inference.

- In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 632–642.
- Pere-Lluís Hugué Cabot and Roberto Navigli. 2021. REBEL: Relation extraction by end-to-end language generation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 2370–2381.
- Marie-Catherine de Marneffe, Anna N. Rafferty, and Christopher D. Manning. 2008. Finding contradictions in text. In *Proceedings of ACL-08: HLT*, pages 1039–1047.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *International Conference on Learning Representations (ICLR)*.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. volume 55, pages 1–38.
- Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2019. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547.
- Deborah L. McGuinness and Frank Van Harmelen. 2004. OWL web ontology language overview. *W3C Recommendation*, 10(10).
- Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. 2020. Adversarial NLI: A new benchmark for natural language understanding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4885–4901.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 809–819.
- Adina Williams, Nikita Nangia, and Samuel Bowman. 2018. A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 1112–1122.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.